MODULARRAGOPTIMIZATION · GEOMETRIC GROUP ANALYSIS

Why Centroid-Based Retrieval Groups Consistently Outperform Query-Centered Groups

RUN	QUERIES	POOL	EXPERIMENTS
staged_run_2026-08-03	196	5,035 docs	3 linked

DATE

2026-08-25

§ 1

Group Definitions

Three geometric groups partition the document pool per query, each defined as all pool documents inside a cosine-distance circle in embedding space.

GROUP	CIRCLE CENTER	RADIUS	MEMBERSHIP RULE
G1	Query vector Q	r_1 = dist(Q, GT)	dist(doc, Q) ≤ r_1
G2	Normalized centroid of G3	r_2 = dist(centroid, GT)	dist(doc, centroid) ≤ r_2
G3	-	-	LLM score = 1.0

Distance = 1 − dot(a, b) for L2-normalised vectors. All DB vectors are unit vectors (HuggingFaceEmbedding normalize=True).

§ 2

Experiment 1 - G1 / G2 / G3 Group Analysis

Ran over all 196 queries using the canonical 5,035-document pool. For each query, G1, G2, and G3 are computed and pairwise overlaps recorded.

GROUP SIZES

G1 - QUERY CIRCLE

3.13

mean docs · median 1.0

G2 - CENTROID CIRCLE

9.72

mean docs · median 6.0

G3 - LLM-SCORED

17.05

mean docs · median 18.0

CENTER–GT DISTANCE

DIST(Q, GT)

0.173

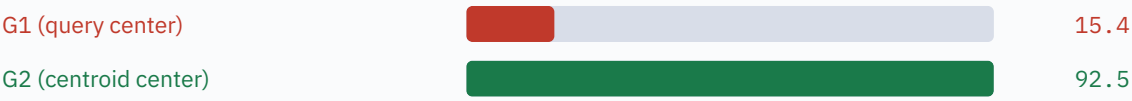
mean cosine distance · median 0.154

DIST(CENTROID, GT)

0.098

mean cosine distance · median 0.086

DENSITY (DOCS PER UNIT RADIUS)



GT MEMBERSHIP AND OVERLAPS

METRIC	G1	G2	G3
GT included (%)	84.7%	85.2%	94.9%
GT in all three (%)			76.0%
Jaccard G1 ∩ G2			0.326
Jaccard G1 ∩ G3			0.125
Jaccard G2 ∩ G3			0.336

Finding: G2 is ~3× larger than G1 (9.72 vs 3.13 docs, mean) in 77.6% of queries, and has 6× higher density (92.5 vs 15.4 docs/radius). The centroid is also 1.77× closer to GT (0.098 vs 0.173). This raises a key question: why?

§ 3

Experiment 2 - Halfpoint Experiment (Random Space Points)

To test whether the GT neighborhood is inherently dense, 10 random unit vectors are generated per query at exactly cosine distance $r_1/2$ from GT. Each anchor's circle has radius $r_1/2$ - same distance-to-GT as a typical centroid, matching G2's convention.

CONSTRUCTION (GRAM-SCHMIDT)

```
u_perp = v − dot(v, GT) · GT      # remove GT component
u_perp /= ‖u_perp‖                # unit vector ⊥ GT
anchor = cos(θ)·GT + sin(θ)·u_perp  # where cos(θ) = 1 − r₁/2
```

Verification: dot(anchor, GT) = cos(θ) → cosine_distance(anchor, GT) = $r_1/2$ ✓

RESULTS - 196 × 10 = 1,960 ANCHORS

GROUP	CENTER	MEAN DOCS	MEDIAN	MEAN DENSITY
G1	Query Q, radius r_1	3.18	1.0	15.6
Halfpoint (HP)	Random anchor, radius $r_1/2$	0.82	1.0	11.0
G2 (reference)	Centroid, radius r_2	**9.78**	**6.0**	-

Queries where HP mean > G1: 7.1% (14/196)

Key result: Random points at the same distance from GT as the centroid capture fewer docs (0.82) than G1 (3.18) - not more. The GT neighborhood is **not inherently dense in arbitrary directions.** The centroid's advantage is not a general property of proximity to GT.

§ 4

Experiment 3 - Pool-Document Centroid Experiment

Instead of synthetic random points, use actual pool documents as circle centers. For each query, find pool docs where dist(doc, GT) < r_1 (closer to GT than the query), sample up to 10, and use each as a center with radius = dist(doc, GT) - identical convention to G2.

This tests whether the pool itself is structurally dense around GT.

RESULTS - 196 QUERIES, 159 WITH ≥1 ELIGIBLE POOL DOC

GROUP	CENTER	MEAN DOCS	MEDIAN	MEAN DENSITY
G1	Query Q, radius r_1	3.18	1.0	15.6
Pool-doc candidate	Real pool doc, radius=dist(doc,GT)	**10.12**	**6.5**	**59.6**
G2 (true centroid)	G3 centroid, radius r_2	9.78	6.0	92.5

Queries where candidate mean > G1: 91.8% (146/159 valid queries)

> **Key result:** Real pool documents closer to GT than Q capture 10.12 docs on average - nearly identical to G2's 9.78. The document pool is **structurally concentrated near GT.** Any pool document near GT works as well as the centroid for raw count; the centroid has the highest density (92.5 vs 59.6) because it minimises total distance to the cluster, yielding the smallest radius.

§ 5

Unified Interpretation

THE QUERY–DOCUMENT GAP

Queries and documents occupy different regions of embedding space. Because the retriever returns GT as the nearest document to Q, almost nothing in the pool is closer to Q than GT - G1 collapses to 1–3 docs on average. Documents near GT, however, are surrounded by other semantically similar documents. The pool clusters around GT in document space.

DENSITY COMPARISON ACROSS ALL EXPERIMENTS

ANCHOR	MEAN DOCS	MEAN RADIUS	DOCS / RADIUS
Query Q (G1)	3.18	0.173	15.6
Random point at $r_1/2$ from GT (HP)	0.82	0.087	11.0
Pool doc closer to GT than Q	10.12	~0.070	59.6
G3 centroid (G2)	**9.78**	**0.098**	**92.5**

The centroid's superiority in density (92.5) vs raw pool docs (59.6) arises because the centroid minimizes total distance to all G3 members - it is the optimal center of the cluster, yielding the tightest radius for the same coverage.

§ 6

Conclusions

1. **G2 systematically outperforms G1**

   In 77.6% of 196 queries, G2 captures more documents (~3× on average: 9.72 vs 3.13). The centroid anchors the circle inside the dense document cluster, not in query space.

2. **The GT neighborhood is not uniformly dense**

   Random points at equal distance from GT capture only 0.82 docs - fewer than G1. The density advantage requires real document structure, not just proximity to GT.

3. **The document pool is structured around GT**

   Real pool docs closer to GT than Q capture 10.12 docs (91.8% of queries beat G1). The pool is not uniform - it concentrates near semantically relevant content.

4. **The query–document gap drives the disparity**

   Queries sit in a different embedding region than documents. Using Q as the circle center misses the document cluster entirely - G1's radius r_1 is "wasted" on empty query space.

5. **The centroid is special in density, not in raw count**

   Any real pool doc near GT gives similar raw count (~10). The centroid has uniquely high density (92.5 vs 59.6) because it minimizes total cluster distance, yielding the smallest radius for the same coverage.

§ 7

Output Files

FILE	DESCRIPTION
`csv/group_analysis/master_summary.csv`	Per-query G1/G2/G3 sizes, overlaps, GT membership (196 rows)
`csv/group_analysis/global_aggregates.csv`	Global mean/median/std for all metrics
`csv/group_analysis/per_query/query_NNN.csv`	Per-doc membership flags, distances, scores per query

FILE	DESCRIPTION
`csv/halfpoint_groups/all_anchors.csv`	1,960 rows (196 × 10 anchors) with HP group sizes and densities
`csv/halfpoint_groups/per_query_summary.csv`	Per-query HP aggregates vs G1 (196 rows)
`csv/halfpoint_groups/global_aggregates.csv`	Global halfpoint statistics
`csv/pooldoc_centroid/all_candidates.csv`	1,052 candidate-doc rows across 159 valid queries
`csv/pooldoc_centroid/per_query_summary.csv`	Per-query pool-doc centroid aggregates (196 rows)
`csv/pooldoc_centroid/global_aggregates.csv`	Global pool-doc centroid statistics

Run: `staged_run_2026-08-03_08-32-45` · Scripts: `analyze_groups.py`, `analyze_halfpoint_groups.py`, `analyze_pooldoc_centroid.py` · Shared data loader: `load_experiment_data.py` · Generated 2026-08-25